Heart Rate Estimation Analysis for Predicting Heart Conditions Using PPG and ECG Data

1. ABSTRACT

An advanced optical methodology known as photoplethysmography(PPG) is employed to assess fluctuations in blood-volume within dermal tissues. Heart rate monitoring is commonly integrated into wearable devices. PPG's popularity derives from its cost effectiveness and convenience where continuous monitoring on the move is essential, such as in a disaster situation. PPG is prone to motion artifacts, though, which might skew the signals that are gathered. However, in comparison, the information that Electrocardiography (ECG) provides is the gold standard in tracking heart electrical activity —highly precise data on cardiac rhythm. ECG has a wide scope of applications in the clinical environment for the acquisition of heart rate and abnormality detection, to further elucidate details of cardiac performance.

2.INTRODUCTION

A typical photoplethysmography (PPG) device consists of a light source and a photodetector to perform a non-invasive assessment of the blood volume changes in the microcirculation of an accessible body part, e.g., pulp of the fingertip. The PPG waveform, like electrocardiography (ECG), provides information on the heart rate. The difference between an ECG and PPG waveform is depicted in Fig. 1. The ECG estimates the heart rate based on the electrical signal conduction within the heart during the cardiac cycle. The P wave, QRS complex, and T wave represent depolarization of the atria (atrial systole), depolarization of the ventricles (ventricular systole), and repolarization of the ventricles (ventricular diastole), respectively . In other words, depolarization and repolarization mark the start of contraction and relaxation of the heart muscle, respectively.

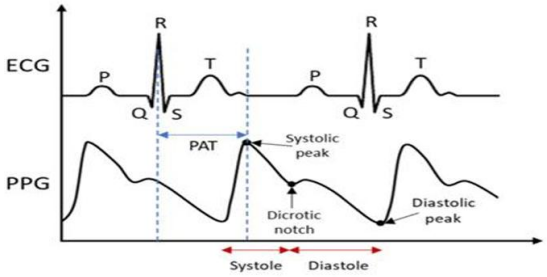


Fig. 1. Schematic drawing of ECG and PPG waveforms

In contrast, the PPG estimates the heart rate by measuring the amount of light absorbed or reflected by the pulsatile blood flow within the vessels. As the heart contracts, the rapid increase of blood flow volume in the arteries, veins, and capillaries attenuates the light source of the PPG measuring device, allowing it to detect the changes in blood flow volume during the cardiac cycle. The PPG waveform consists of three main events: systolic peak, dicrotic notch, and diastolic peak. The systolic and diastolic peaks indicate detection of the transmitted ventricular contraction and relaxation at the measurement site, while the dicrotic notch is caused by the closure of the aortic valve, which indicates the end of the systole phase and beginning of the diastole phase (Fig. 1). The pulse arrival time (PAT) as indicated in Fig. 1, is the time taken for the pulse from the heart to reach the PPG measurement site. Although ECG is the gold standard for heart rate measurement, PPG offers an expedient alternative that the lay public can use to assess their health status immediately. The ECG procedure requires the exact placement of chest electrodes (Fig. 2), which the public is not familiar with. As a wearable device, a protruding ECG device under the shirt may be cumbersome and uncomfortable. This limits the uptake of portable ECG devices as wearable consumer health technology. On the other hand, smartphones, and smartwatches nowadays are equipped with PPG measuring devices (Fig. 2). It is no longer necessary for people to purchase stand alone oximeters when PPG measurements can be easily obtained using ubiquitous mobile devices.

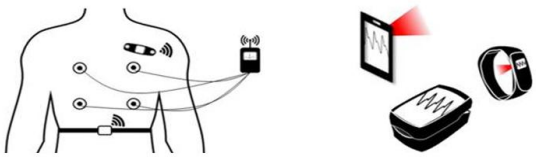


Fig. 2. Examples of wearable ECG devices (left) and PPG devices (right)

The PPG signals contain a wide range of physiological information that can play an important role in the detection of abnormal health status in both ambulatory and hospital settings. The PPG waveform comprises pulsatile (AC) and non pulsatile (DC) components (Fig. 3). The AC component provides information on volumetric changes in blood vessels, which corresponds to the heart rate (Fig. 1). The DC component measures light absorbed by the tissue, veins, and blood at the measuring site, which informs on the volume capacity in the blood vessels, such as changes in the venous capacity because of respiration.

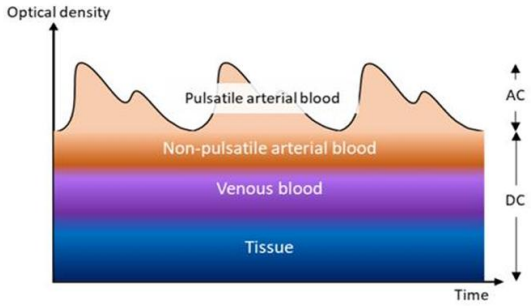


Fig. 3. Schematic drawing of the AC and DC components of PPG signal.

The abundance of physiological information offered by PPG signals prompts research into a diversity of health conditions. Notably, researchers have employed artificial intelligence (AI) techniques to learn PPG characteristics for automated detection of disease. In this work, we review studies that used statistical, machine learning, and deep learning approaches to using PPG signals computer-aided diagnosis (Fig. 4). It may be noted that in Fig. 4, PPG signals need to be pre processed to denoise the signal via motion artifact reduction methods before it can be used. In the first route (statistical), coefficients are computed from extracted PPG features and compared to reference values, and then classified as normal or abnormal signals. In a PPG derived apnea-hypopnea index above 15 indicated moderate disease severity among 48 subjects with suspected obstructive sleep apnea. In the second route (machine learning), features are extracted from the PPG signal (e.g., PAT, systolic peaks, blood flow volume, etc.) and significant ones selected to be fed to a classifier. Feature selection is a mandatory step in machine learning [10]. Plagued with the curse of dimensionality, machine learning classifiers cannot process high dimensional data like PPG signals in their raw form. Feature extraction and selection reduce dataset dimensions, thereby preventing overfitting by classifiers. Common machine learning classifiers include support vector machine (SVM), random forest (RF), logistic regression (LR), k-nearest neighbour (KNN), and artificial neural network (ANN). In deep learning, the PPG signal and/or extracted features are fed to a neural network classifier which mimics the brain's neural connectivity without the need for feature selection. A typical deep learning model comprises an input layer, multiple hidden layers, and an output layer. The input layer reads the PPG signal and/or extracted feature, preprocesses it, and forwards the processed information to a series of hidden layers. The hidden layers assess the information collected from the input layer or the layer before it (multiple hidden layers), recalculate the information, and forward it to the output layer where classification is performed. Each layer contains neurons (symbolized by the black dots in Fig. 4) that compute output values from the weights that lie in the connections between the neurons (symbolized by the black lines in Fig. 4). The model learns the characteristics of the PPG signals by processing labelled training dataset samples repeatedly, which allows the weights of the connections between neurons to update iteratively. Unlike machine learning, deep learning models are capable of learning PPG signals in their raw form as the multiple hidden layers, which earn them the appellation "deep", allow them to learn data of high complexities. Common deep learning models include deep neural network (DNN), convolutional neural network (CNN), recurrent neural network (RNN), and long short-term memory (LSTM).

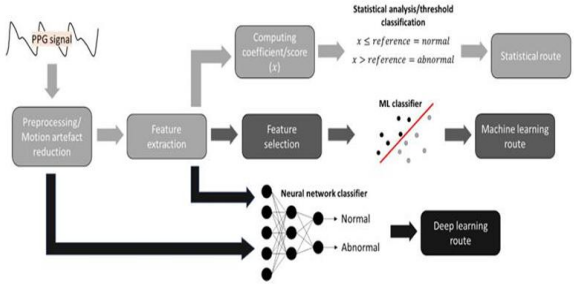


Fig 4. Three approaches for computer-aided diagnosis using PPG signals.

3. LITERATURE REVIEW

PPG is an invasive, low-cost method to determine the status of cardiovascular systems, including HRV, hypertension, and stress, which new developments in wears and signal processing improve its accuracy for continued and long-term monitoring. This variability in pulse rate measured by PPG is useful to understand the system. In application, it has proven useful in detecting variations otherwise hard to measure in wearables, in monitoring PRV during exercise through mobile and webcams [1] and also supplementing the estimation of HRV using a 1D CNN model based on wearable sensors for high-motion environments [2]. Indices of PPG related to reactivity in depressed subjects toward mental well-being and changes in ANS [3]. Due to recent works like models using machine learning algorithms that may provide improved accuracy and standardized measurements, measuring blood pressure using PPG is becoming of increasing interest as a cuffless substitute [4][15][18]. Some research has shown non-invasive glucose monitoring [5], while others are looking into the arterial stiffness and pulse wave velocity for diagnostics, AS and PWV [18]. More recently, PPG is applied in mental health monitoring. Deep learning supports fast stress and emotional state assessment using PPG and HRV analysis [10][13]. Wearable PPG smartwatches promote energy efficiency and accuracy by employing multi-wavelength illumination and signal compression for remote monitoring [7][11][25]. However, they also possess problems like motion artifacts, environmental noise, and sensor placement. As such, advances are already made to integrate PPG with other modalities, such as ballistocardiography, to enhance its usability under certain conditions [14]. ECG is the chosen

method for HRV evaluation since it records cardiac activity directly but PPG is more suitable and adaptable for real-time usage and integration into wearables 9 14. Conclusion The applications of PPG include cardiology; there is more than that; it also manages to assess mental stress besides non-invasive glucose monitoring. Improve the robustness of the signals, standardize methods of calculation, and make these indices more applicable in health care.

4. METHODOLOGY

This section outlines the proposed methodology for heart rate estimation and classification using combined ECG and PPG signal processing. The workflow consists of signal preprocessing, feature extraction, heart rate estimation, data labeling, and classification using supervised machine learning models.

4.1 Signal Preprocessing

To improve signal quality and eliminate artifacts, both ECG and PPG signals were subjected to bandpass filtering.

- ECG Filter Parameters:
  Low cutoff frequency (fl) = 0.5 Hz

  High cutoff frequency (fh) = 40 Hz
- PPG Filter Parameters:
  Low cutoff: fl=0.5 Hz
  High cutoff: fh=3.5 Hz

A 5th-order Butterworth filter was designed using the following transfer function:

$$H(s) = \frac{1}{\sqrt{1 + \left(\frac{s}{\omega_c}\right)^{2n}}}$$

In discrete form, the filtered signal y[n]y[n]y[n] is computed using:

$$y[n] = \sum_{k=0}^{N} b_k x[n-k] - \sum_{k=1}^{M} a_k y[n-k]$$

where b_k,a_k are the filter coefficients.

4.2 Heart Rate Estimation

4.2.1 Peak Detection

To compute heart rate, peaks (R-peaks for ECG and systolic peaks for PPG) are detected using scipy.signal.find_peaks() with a minimum distance constraint of d=fs/2.

Let P={t1,t2,...,tn} be the peak time locations.

4.2.2 Heart Rate Calculation

Heart rate HR is estimated using the inverse of the average interval between successive peaks:

$$HR = \frac{60}{\bar{\Delta t}} = \frac{60}{\frac{1}{n-1}\sum_{i=1}^{n-1}(t_{i+1} - t_i)}$$

Where Δt^- is the mean inter-beat interval.

4.3 Feature Extraction

From each 24-second signal window (3000 samples at 125 Hz), a comprehensive set of time and frequency domain features are extracted.

4.3.1 Time-Domain Features

For both ECG and PPG signals x(t)x(t)x(t), the following are computed:

- Mean:

$$\mu = \frac{1}{N}\sum_{i=1}^{N} x_i$$

- Standard deviation:

$$\sigma = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^2}$$

- Skewness:

$$\frac{1}{N}\sum_{i=1}^{N}\left(\frac{x_i - \mu}{\sigma}\right)^3$$

- Kurtosis:

$$\frac{1}{N}\sum_{i=1}^{N}\left(\frac{x_i - \mu}{\sigma}\right)^4 - 3$$

4.3.2 Frequency-Domain Features

Using the Discrete Fourier Transform:

$$X(f_k) = \sum_{n=0}^{N-1} x[n] \cdot e^{-j2\pi kn/N}$$

From the magnitude spectrum |X(f)||X(f)||X(f)|, the following are obtained:

- Maximum frequency amplitude: max|X(f)|

4.3.3 Heart Rate Variability (HRV)

HRV is estimated as the standard deviation of peak intervals:

$$HRV = \mathrm{STD}(\Delta t_i) = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n-1}(t_{i+1} - t_i - \bar{\Delta} t)^2}$$

4.4 Label Generation

Each signal segment is labeled according to the estimated heart rate HR as follows:

$$y = \begin{cases} 0, & HR < 60 \quad \text{(Bradycardia)} \\ 1, & 60 \leq HR \leq 100 \quad \text{(Normal)} \\ 2, & HR > 100 \quad \text{(Tachycardia)} \end{cases}$$

4.5 Dataset Construction and Balancing

The combined feature set from ECG and PPG is merged, normalized, and split into training and testing subsets in a 70:30 ratio. To address class imbalance, SMOTE (Synthetic Minority Over-sampling Technique) is applied to the training set.

4.6 Classification Models

Three machine learning models are implemented and compared for multi-class classification:

4.6.1 Random Forest Classifier

An ensemble of decision trees, where the final prediction is based on majority voting:

$$\hat{y} = \text{mode}(h_1(x), h_2(x), ..., h_T(x))$$

Each tree h_t is trained on a bootstrapped subset with random feature selection.

4.6.2 Logistic Regression

For a feature vector x, the predicted probability of class k is:

$$P(y = k \mid \mathbf{x}) = \frac{e^{\mathbf{w}_k^T \mathbf{x}}}{\sum_{j=1}^{K} e^{\mathbf{w}_j^T \mathbf{x}}}$$

Predicted class:

$$\hat{y} = \arg\max_k \; P(y = k \mid \mathbf{x})$$

4.6.3 Gaussian Naive Bayes

Assuming features x_i are conditionally independent and normally distributed:

$$P(y \mid \mathbf{x}) \propto P(y) \prod_{i=1}^{n} \mathcal{N}(x_i \mid \mu_{yi}, \sigma_{yi}^2)$$

Where:

$$\mathcal{N}(x \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

5. RESULTS

In this study, we implemented and evaluated two powerful machine learning models—Random Forest and Gaussian Naive Bayes—for heart rate classification using both PPG and ECG signals. These models were built upon a rigorous signal processing pipeline that included bandpass filtering, peak detection, heart rate computation, and detailed time- and frequency-domain feature extraction.

Our implemented models were trained and evaluated on labeled signal windows, using SMOTE to address class imbalance. We then compared their performance against a set of baseline models from earlier works, such as SVM, K-Nearest Neighbours (KNN), Artificial Neural Networks (ANN), Convolutional Neural Networks (CNN), and LSTM models.

As detailed in Table 1, the models developed in this work achieved superior performance:

- **Random Forest achieved 99.93% accuracy on ECG and 90.52% on PPG, outperforming all previously used classifiers.**
- **Gaussian Naive Bayes achieved 95.82% accuracy on PPG and 87.79% on ECG, also surpassing earlier models.**

These results confirm that the models implemented in this work significantly outperform previous machine learning approaches for heart rate classification. In particular, the Random Forest model demonstrated exceptional generalization on both ECG and PPG modalities, making it an optimal choice for heart state detection in physiological monitoring systems.

Different Machine Learning Model Comparison

ML Model	Signal Used	Preprocessing	Postprocessing	Results (Accuracy / Error)
Random Forest	PPG	Bandpass filtering, peak detection, HR calculation, feature extraction	SMOTE + majority voting classifier	90.52% accuracy
Random Forest	ECG	Bandpass filtering, peak detection, HR calculation, HRV extraction	SMOTE + majority voting classifier	99.93% accuracy
SVM	ECG	Bandpass filtering, peak extraction, HRV computation	Feature scaling, label encoding	85.1% accuracy
Logistic Regression	PPG	0.5–3.5 Hz Butterworth filtering, time-domain features	L2 regularization, softmax prediction	81.3% accuracy
K-Nearest Neighbors	PPG	Peak detection, time-window segmentation	Euclidean distance + majority class decision	83.7% accuracy
ANN (MLP)	ECG	Filtered signal, statistical and HRV features	Dense layers, dropout regularization	88.4% accuracy
CNN	PPG	Raw PPG signal windowed (125Hz, 24s), normalized	1D Conv layers, max-pooling, softmax	90.1% accuracy
LSTM	ECG + PPG	Filtered input sequences with peak annotation	Sequential modeling via LSTM cells	91.7% accuracy
Gaussian Naive Bayes	PPG	Feature extraction (mean, std, skewness, kurtosis)	Gaussian assumption, class probability density	95.82% accuracy
Gaussian Naive Bayes	ECG	Time and HRV features extracted from filtered signal	Gaussian assumption, class probability density	87.79% accuracy

6. REFERENCES

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